

HoloArm: A Face-Following 3D Display Using Autostereoscopic Display and Robot Arm

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Figure 1: Our *HoloArm* concept allows for a unique autostereoscopic experience at various viewing angles. The viewer can (a) face forward, (b) move 90 degrees clockwise to look up, or (c) move 120 degrees clockwise to look down

ABSTRACT

This work introduces *HoloArm*, an actuated autostereoscopic display that consistently maintains itself in front of the user. This system balances an actual-size presentation of stereoscopic 3D contents with quick real-virtual context switch capabilities. Three-dimensional displays are known to enhance users' spatial awareness and immersion in content. However, the predominant head-mounted displays obstruct users' view of the real world and necessitates cumbersome donning and doffing. And autostereoscopic displays allow 3D viewing without eye-glasses, but are constrained by a limited field of view and display size. To address these challenges, *HoloArm* obviates the need for users to wear additional equipment and dynamically adjusts the display according to the user's face position and orientation, achieving a broad field of view and large image display. We have implemented a prototype of *HoloArm* and evaluated its tracking latency.

Index Terms: Human-centered computing Visualization; Human-centered computing Mixed / augmented reality

1 INTRODUCTION AND BACKGROUND

The term 3D displays encompass a variety of technologies based on where they spatially form images [1]. Examples include CAVE systems [8], autostereoscopic displays [3, 6], and head-mounted displays (HMDs) [2, 5]. In particular, recent years have witnessed remarkable advancements in HMDs, particularly within VR and AR.

However, HMDs come with their own set of challenges. Their weight remains a substantial impediment [4], making them unsuitable for extended usage or activities requiring mobility. Furthermore, obstructing the user's real-world view poses a significant concern, hampering interactions with the physical environment while wearing HMDs. And autostereoscopic displays and holograms also present a

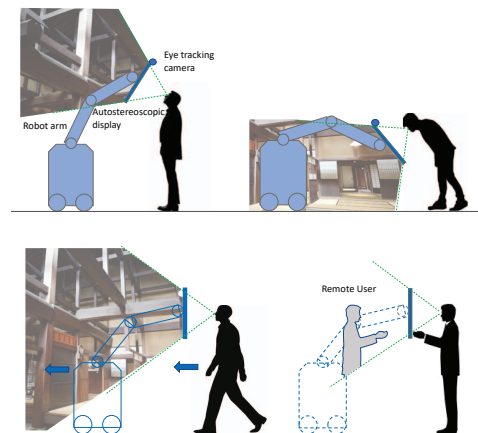


Figure 2: A schematic illustration of our *HoloArm* concept: (Above) The autostereoscopic display always moves so that it is directly opposite the user's head. (Bottom left) The entire display system moves with the user's movements. (Bottom right) Life-size communication with a remote user is also possible.

separate challenge of limited field of view and size constraints that limit the degree of freedom of viewpoint adjustment.

Therefore, we propose a new paradigm, *HoloArm*, that entails mounting an autostereoscopic display onto a robotic arm, which then autonomously tracks and maintain itself in front of the user's eye while keeping its functionalities. *HoloArm* aims to give users a wider perspective on 3D images than conventional autostereoscopic displays, addressing the challenges posed by HMDs while achieving an equivalent or superior field of view (Fig. 1).

In this work, we implemented a prototype of *HoloArm*, employing a robotic arm and a 27-inch autostereoscopic display for our initial demonstration.

2 CONCEPT

To realize the *HoloArm* concept shown in Fig. 2, we propose a system that includes an autostereoscopic display and a robotic arm. Fig. 3 shows the system configuration of the *HoloArm*. The au-

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Table 1: Tracking delay results.

Motion type	Time from arm movement to stop [s]	Until the user perceives binocular images[s]
1m from the front	30.27s	18.73s
Up and down 1m	23.80s	15.47s
90 degree arc	34.6s	10.5s

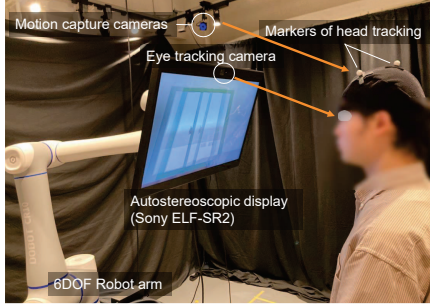


Figure 3: Capture of our HoloArm prototype configuration. A robot arm installed on the base platform is holding an autostereoscopic display. The display tracks the user's eye position. The motion capture system on the ceiling takes care of robust head pose tracking for robot arm control.

tostereoscopic display is attached to the robot arm, and the camera of the autostereoscopic display detects the users' 3D eye positions. The autostereoscopic display's constantly renders the image according to the detected 3D eye positions and display's own position and rotation. The movement of the human head is tracked by an infrared marker cap worn by the user. When the user moves, the movement is transmitted to the robotic arm via the infrared marker, and the robotic arm moves to follow the human head. The robot follows the position detected by the infrared marker. These two tracking methods capture the user and change the displayed image as the user moves. The user can experience the sensation that the world on the display changes as the user moves and can have a natural virtual reality experience. The advantage of this configuration is that the rendering of the 3D image continues to work as long as the user is within the tracking range of the autostereoscopic display, even if there is a delay in following the user's head by the robotic arm.

3 SYSTEM ARCHITECTURE

We have implemented a prototype to demonstrate the concept of the HoloArm (Fig. 3). In this prototype, a robotic arm is fixed to the base, and an autostereoscopic display is attached to the end of the arm with a standard VESA mount. For the tracking stability of the prototype, retro-reflective markers on the user's head is being tracked via OptiTrack system. The 6DoF information of the user's head rotation is sent to the system, which in turn computes the desired end effector position of the arm. The robotic arm moves the display to follow the line of sight based on the transmitted information. The image displayed by the display moves by the displacement of the display.

4 EVALUATION

In this section, we conduct a preliminary experiment to measure the tracking delay of the prototype. We allowed the prototype to track the user and measured the tracking time. First, the robot was placed in front of the user, and the user moved 1 m horizontally, 1 m vertically, and 90 degrees (1.2 m radius) around the robot arm. The user then turned on the robot's tracking mode and measured the time from when the robot moved and stopped until the user recognised

the stereoscopic view. 2 measurements were taken and averaged. The results are summarised in the table 1

On the other hand, these results should be interpreted with caution. While the absolute latency is of the order of 10 seconds, the relative recovery time is short thanks to eye tracking. The absolute latency is due to the use of the joint base of the robot arm as a control for tracking stability. The main implication of this result is that even if the arm's own tracking is delayed, the actual delay time can be perceptually reduced by 40-70 percent due to eye tracking on the autostereoscopic side and compensation by the stereo display.

5 DISCUSSION

This section discusses the potential applications of this prototype while investigating ways to reduce the tracking delay of the current prototype.

5.1 Reducing the Tracking Delay

At this stage, the HoloArm does not compute a pointing path to the user and is driven solely by the joint control and translation/rotation of the arm coordinates on Unity3D. Although some processing is incorporated to avoid inverse kinematics and collision errors, the problem is that the system sometimes chooses a circuitous path to follow the user. For faster tracking and practical control, pathfinder control should be implemented instead of joint control, or the two should be combined.

5.2 Potential Applications

If the concept of presenting life-size 3D images and facilitating the transition between the real and virtual worlds is realized, there are many possible applications.

Firstly, the system offers a new 3D modeling experience: modeling with an HMD does not allow the user to see the keyboard and specifications in the real world; with the HoloArm, the user sees an immersive, life-size 3D image in their eyes without any additional equipment. This allows engineers to create models exactly as they imagine them, without having to put on and take off the HMD. This can also be applied to use cases such as remote robot manipulation. Holoarm also has potential as a telepresence system. For example, the interior of a house could be imaged by NeRF [7] and photographed by HoloArm, which could then be used to preview new homes. This also could be used in museums and art galleries, for example. In this way, HoloArm is well suited to telepresence, allowing people to experience a place where they cannot actually be realistically. Other possibilities include presenting 3D moving images for motion support and designing games by adding new interactions. If the background is darkened, the distinction between the display and the real world becomes indistinguishable, so it could be used as a spatial AR application that seamlessly blends the real and virtual worlds.

6 CONCLUSION

In this study, we introduced HoloArm, an actuated autostereoscopic display system that addresses the limitations of current 3D display technologies by moving the display along the user's face position. While there are challenges ahead, HoloArm represents a significant advancement in delivering adaptable, immersive 3D experiences without the need for wearables. Future work will focus on system miniaturization and broader user studies to validate its effectiveness.

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